Linear Algebra

AshaLy

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Version 0.1.0

Author
Muho Choi
Email: sihyeon8240@gmail.com
GitHub: https://github.com/sihyeon8240

Source Code & Contributions
GitHub: https://github.com/sihyeon8240/AshaLy
Feel free to open issues or pull requests for corrections.

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Preface

This textbook uses Lean 4 to formalize selected mathematical statements and proofs.

- **Tools.** Lean is an interactive theorem prover; Mathlib is a library of formalized mathematics. This edition uses Lean 4.35.0 and Mathlib v4.35.0 (revision `56a005855ea1`).
- **Reading the proofs.** Each paired statement or proof step gives a mathematical explanation followed by its Lean code. The explanations can be read independently; code fragments are read in order within the complete declaration.
- **Verification.** Lean checks formal proofs under their stated assumptions. Verification covers only statements linked to checked declarations. Agreement between the explanations, displayed code, and checked proofs is reviewed separately.

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Bibliography

[1] Serge Lang, *Linear Algebra*, Springer Nature.